Krushnat Kapse — Personal & Professional Profile

A snapshot based on information shared across our conversations

1. Professional Background

Krushnat Kapse completed a B.Tech in Computer Engineering from N. B. Navale Sinhgad College of Engineering, Solapur, with a CGPA of 9.86 during 2019–2023.

He has been working as a System Engineer at Tata Consultancy Services in Pune since October 2023. Earlier, he worked as a Software Engineer Intern at Persistent Systems from June to September 2023.

2. Technical Experience & Interests

- Python, Java, PySpark and SQL
- AWS and Azure DevOps
- Jenkins, CI/CD, GitHub Actions, Git and Linux
- Django and automation
- AWS services including EC2, S3, Lambda, Glue, Athena, API Gateway, CodePipeline, CloudWatch, CloudFormation, ECS, SNS, RDS, Kendra and Fargate
- SAS-to-PySpark migration and data engineering workflows

3. Current AI/ML Learning Direction

Krushnat is building toward practical AI and agentic AI, while deliberately developing enough machine-learning fundamentals to maintain a strong conceptual base.

- Refreshing Python and progressing through NumPy, Pandas and statistics
- Building a foundation in machine learning and later deep learning
- Learning LLMs, RAG, embeddings and vector databases
- Progressing toward AI agents and LangGraph
- Exploring MCP, A2A, A2UI, agent CLIs, ambient agents and spec-driven development
- Using Google Colab as a practical learning environment

4. Current RAG Learning Project

A major recent focus has been understanding RAG from the ground up and clearly separating each component's responsibility.

- PDF → PyPDFLoader → LangChain Documents
- Documents → RecursiveCharacterTextSplitter → chunks
- Chunks → SentenceTransformer → embeddings
- Embeddings + text + metadata → ChromaDB
- User question → query embedding → ChromaDB similarity search
- Retrieved chunks → context → local Ollama LLM → final answer

5. Local AI Setup

Krushnat is experimenting with Ollama for local LLM inference. The local environment includes llama3:latest (8B, Q4_0, about 4.7 GB) and qwen2.5:3b (3.1B, Q4_K_M, about 1.9 GB). Qwen 2.5 3B is currently useful for lightweight RAG development, while Llama 3 is available as a larger alternative.

6. Capital Markets / Compliance Work

Krushnat has worked on an AI-oriented capital-markets compliance application involving ingestion, validation, enrichment, SQLite, RAG and transaction-level audit trails. The audit concept focused on tracing transactions by identifiers rather than simply auditing the application itself.

7. Learning Style

- Prefers simple explanations first, followed by progressively deeper detail.
- Wants to understand what components actually do before relying on frameworks.
- Uses precise follow-up questions to validate data flow and function behavior.
- Prefers concise answers for straightforward questions and deeper explanations for technical learning.
- Focuses on learning by building practical projects.

8. Recent RAG Milestone

A key conceptual milestone has been separating the roles of SentenceTransformer, ChromaDB and the LLM: SentenceTransformer creates embeddings; ChromaDB stores and retrieves vectors with associated document data; and the LLM generates the final answer from retrieved context.

This is a snapshot of relevant professional, technical and learning information available from prior conversations—not a complete biography.